

Regular/Supplementary Summer Examination – 2025

Semester : V

Duration: 3 Hr.

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)			12
1 Which renewable energy source uses the Earth's internal heat?		CO1	1
a) Geothermal b) Biomass c) Solar d) Tidal			
2 Which of the following is not a type of primary resource?		CO1	1
a) Crude Oil b) Coal c) Hydrogen d) Sunlight			
Energy			
3 Which of the following states is known oil reserves?		CO1	1
a) Jharkhand b) Odisha c) Madhya d) Assam			
Pradesh			
4 Evacuated flat-plate solar collectors are a type of _____		CO3	1
a) b) c) non- d) solar stills			
concentrating photovoltaic concentrating			
collectors technology collectors			
5 What is solar irradiance?		CO2	1
a) Solar b) Solar c) Solar d) Intensity			
radiation radiation radiation with which the			
received by directly received by solar radiation			
earth received by earth's surface enters the			
earth's surface after scattering earth's			
			atmosphere

6	Which of the following is generally used to measure direct solar radiation? a) Pyranometer b) Actinometer c) Pyr heliometer d) IC tester	CO2	1
7	Which of the following is used to produce biogas from biomass? a) Anaerobic b) Aerobic c) Fermentation d) Pyrolysis treatment treatment	CO4	1
8	Which of the following is a mechanical application of wind energy? a) Lighting b) Electricity c) Pumping d) Wind water surfing	CO4	1
9	Which of the following is a part of a general wind energy conversion system? a) Server b) Wind c) Cloud User d) Service turbine Provider	CO4	1
10	How does a tidal lagoon work? a) By storing b) By creating c) By placing d) By creating sea water a difference in the turbine in barrages water levels the running water	CO5	1
11	_____ turbine is used in closed cycle ocean thermal energy conversion. https://www.batuonline.com a) Horizontal b) Low- c) High- d) Vertical pressure pressure	CO5	1
12	Which of the following is used as a moderator in a nuclear reactor? a) Cadmium b) Plutonium c) Uranium d) Heavy water	CO5	1
Q. 2 Solve the following.			12
A)	Differentiate between Renewable Energy sources and non-Renewable Energy sources.	CO1	6
B)	Explain in detail energy reserves in India.	CO1	6
Q.3 Solve the following.			12
A)	Define a) Declination angle b) Zenith angle c) Altitude angle.	CO2	6

B)	Explain with neat sketch construction and working of Flat plate collector. Also write its advantages and disadvantages.	CO2	6
Q. 4	Solve Any Two of the following.		12
A)	What is solar still? Explain solar distillation system with neat sketch.	CO3	6
B)	Explain any two types of solar concentrating collectors with neat sketch.	CO3	6
C)	What is PV cell? Explain how it converts solar energy into electrical energy.	CO3	6
Q.5	Solve Any Two of the following.		12
A)	What is wind energy? Explain selection of site for wind turbine generation system.	CO4	6
B)	How the wind mills are classified? Sketch the diagram of a Horizontal Axis Wind mill, and explain the function of its main components.	CO4	6
C)	What is biomass energy? Explain with neat sketch floating drum type Biogas plant.	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	What is OTEC system? Explain with neat sketch the working of open cycle OTEC system.	CO5	6
B)	Write a short note on 'Geothermal energy'.	CO5	6
C)	Draw neat sketch of Hydro-electric power plant. How the electric energy is generated in a hydroelectric power plant?	CO5	6

*** End ***